# IGCSE 0607 Practice Questions — E10.8 — Combined Questions & Markscheme

## Cumulative frequency diagrams

**Total Questions:** 9 — **Total Marks:** 68 Marks

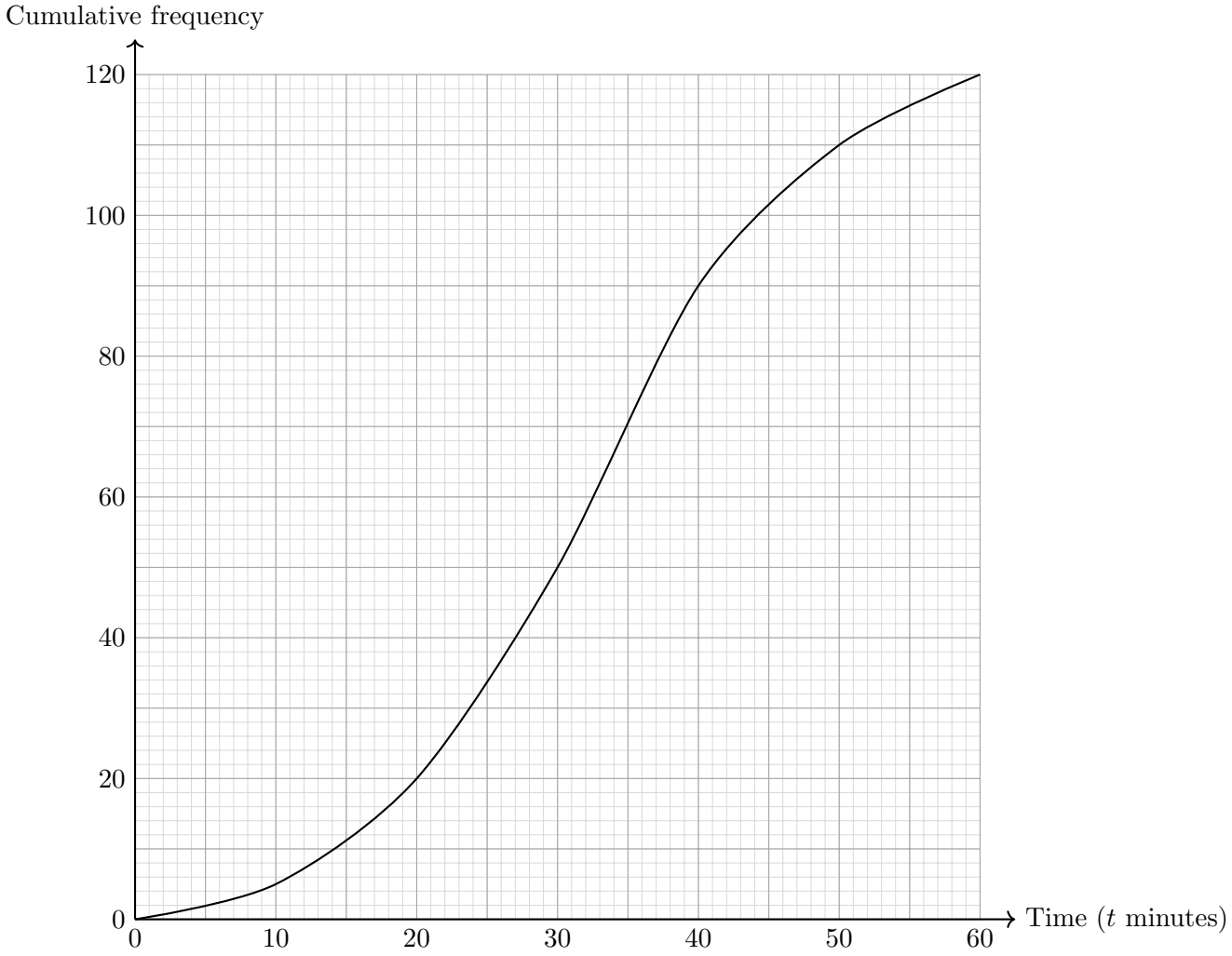
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**1** The cumulative frequency diagram shows the times taken, $t$ minutes, by 120 students to complete a test.




**(a)** Use the diagram to estimate the median time.

..... minutes **[1]**

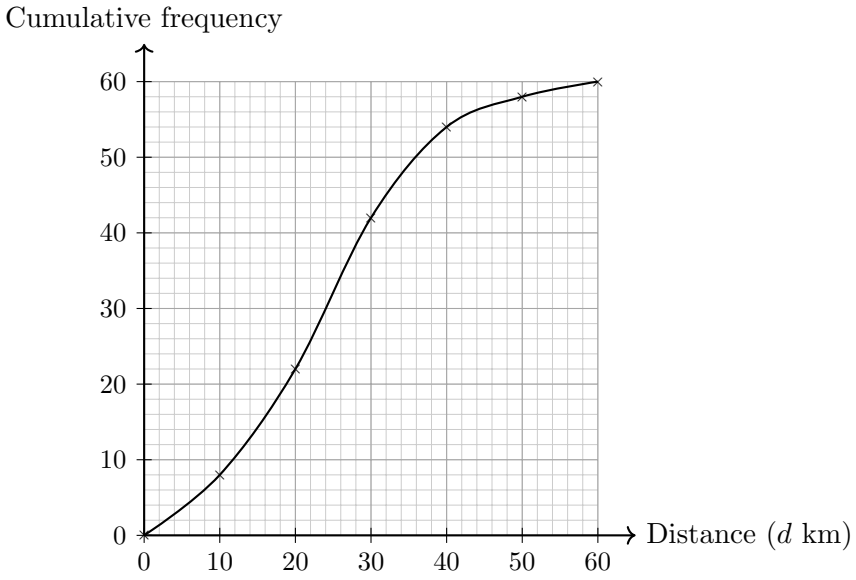
**(b)** Use the diagram to estimate the interquartile range.

..... minutes **[2]**

**(c)** Students who take longer than 45 minutes are given a penalty.
Estimate the number of students who are given a penalty.

..... **[2]**

**2** The cumulative frequency diagram shows information about the distance, $d$ km, travelled by 60 people to work.




**(a)** Use the diagram to find an estimate for:

**(i)** the median,

..... km **[1]**

**(ii)** the interquartile range.

..... km **[2]**

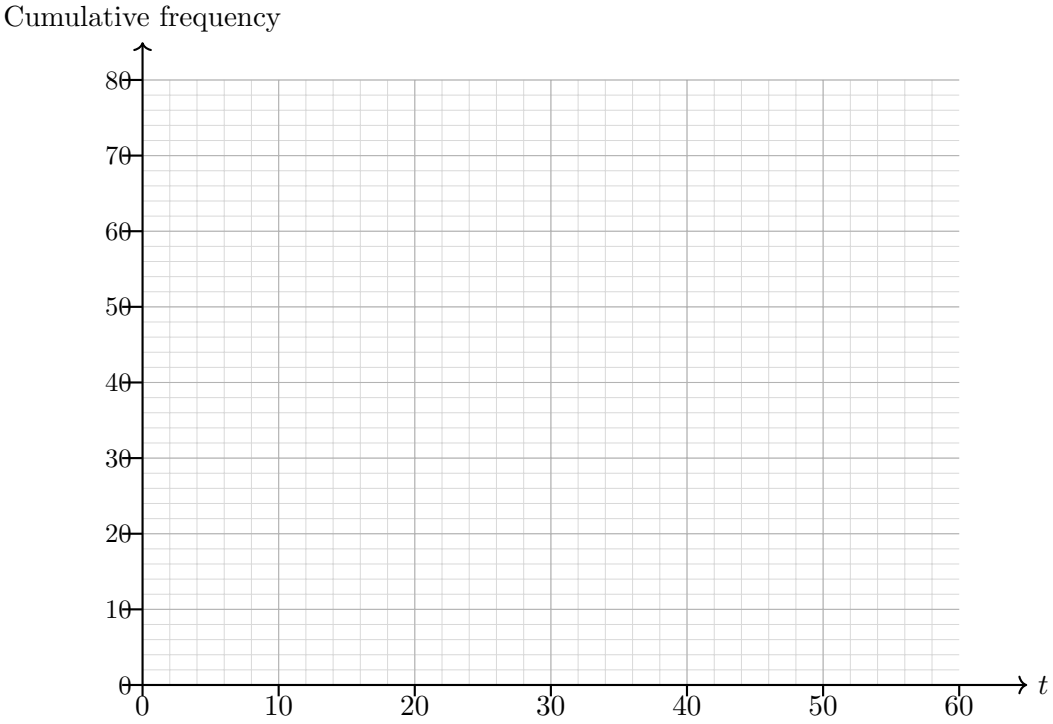
**(b)** Find the number of people who travelled more than 45 km to work.

..... **[3]**

**3** The table shows the cumulative frequencies for the times, $t$ minutes, taken by 80 students to complete a puzzle.

| Time ($t$ minutes) | $t \leq 10$ | $t \leq 20$ | $t \leq 30$ | $t \leq 40$ | $t \leq 50$ | $t \leq 60$ |
|---|---|---|---|---|---|---|
| Cumulative frequency | 6 | 22 | 48 | 68 | 78 | 80 |

**(a)** On the grid, draw a cumulative frequency diagram to show this information.




**[3]**

**(b)** Use your diagram to estimate the median time.

..... minutes **[2]**

**4** A company records the salaries, $s$ thousand dollars, of its 200 employees.
The frequency table shows the results.

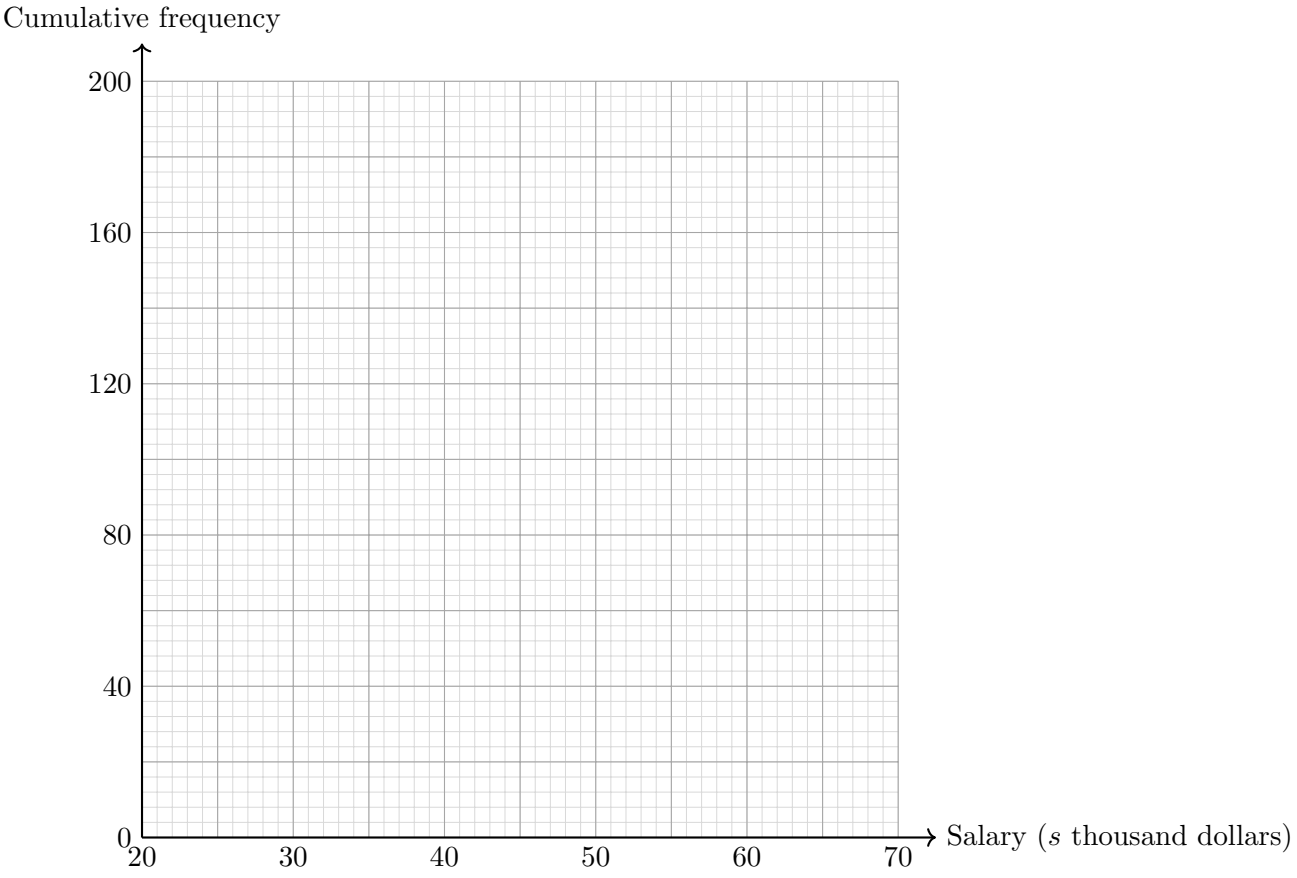
| Salary ($s$ thousand dollars) | $20 < s \leq 30$ | $30 < s \leq 40$ | $40 < s \leq 50$ | $50 < s \leq 60$ | $60 < s \leq 70$ |
|---|---|---|---|---|---|
| Frequency | 24 | 66 | 70 | 28 | 12 |

**(a)** Complete the cumulative frequency table.

| Salary ($s$ thousand dollars) | $s \leq 30$ | $s \leq 40$ | $s \leq 50$ | $s \leq 60$ | $s \leq 70$ |
|---|---|---|---|---|---|
| Cumulative frequency | 24 | | | | |

**[2]**

**(b)** On the grid, draw a cumulative frequency diagram to show this information.

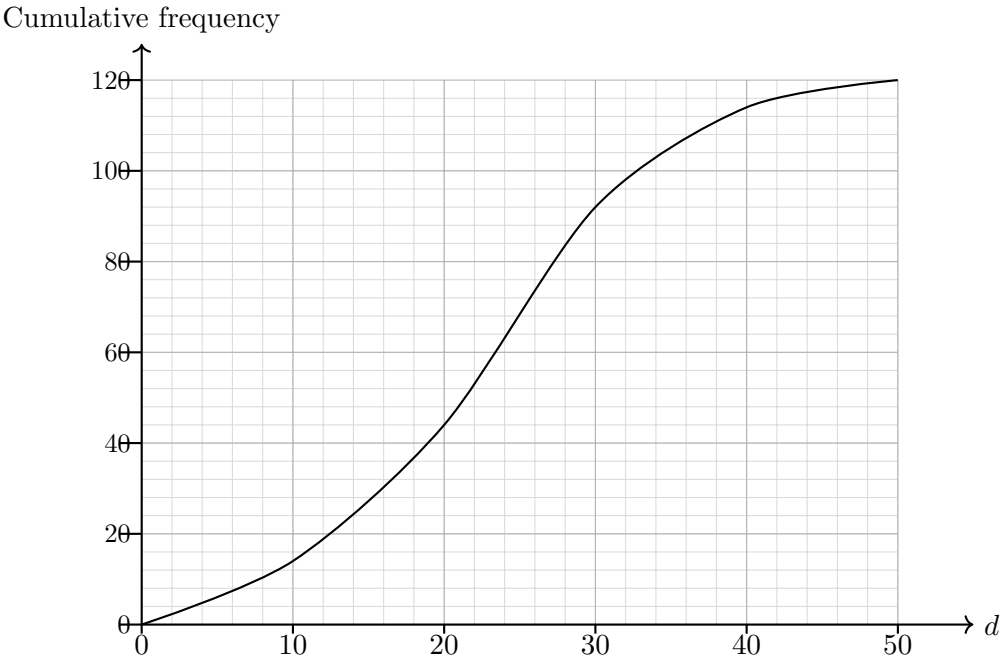



**[3]**

**(c)** Use your diagram to estimate the 70th percentile of the salaries.

..... thousand **[2]**

**5** The cumulative frequency diagram shows information about the distances, $d$ km, travelled by 120 people to work.




**(a)** Use the diagram to estimate the median distance.

..... km **[2]**

**(b)** Use the diagram to estimate the interquartile range.

..... km **[3]**

**(c)** Estimate the number of people who travelled more than 35 km to work.

..... **[3]**

**6** A table shows information about the masses of 120 apples.

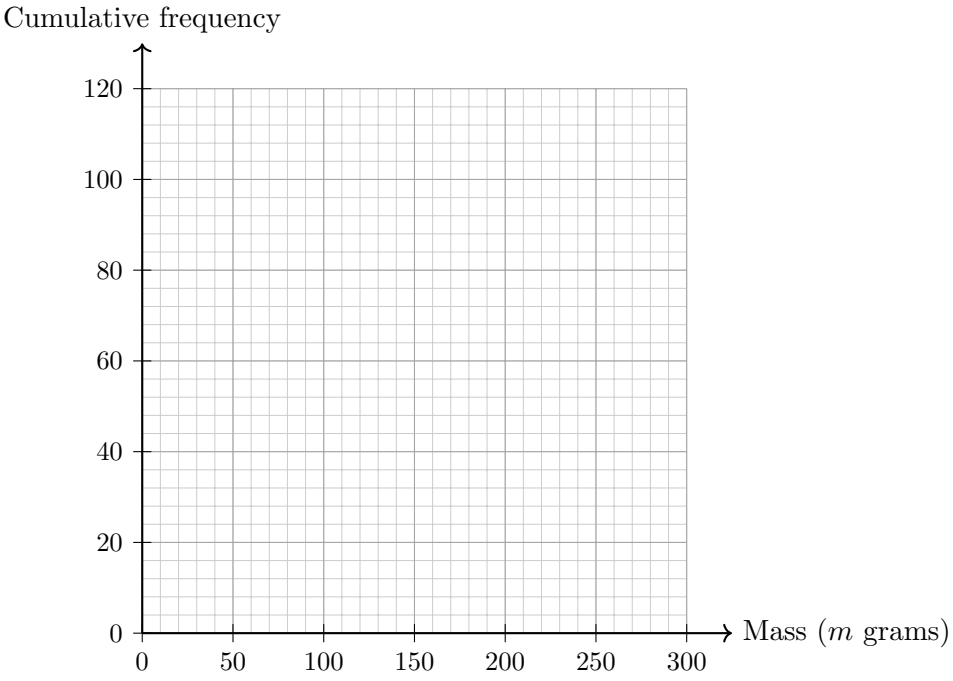
| Mass ($m$ grams) | $50 < m \leq 100$ | $100 < m \leq 150$ | $150 < m \leq 200$ | $200 < m \leq 250$ | $250 < m \leq 300$ |
|---|---|---|---|---|---|
| Frequency | 12 | 34 | 46 | 20 | 8 |

**(a)** Complete the cumulative frequency table.

| Mass ($m$ grams) | $m \leq 100$ | $m \leq 150$ | $m \leq 200$ | $m \leq 250$ | $m \leq 300$ |
|---|---|---|---|---|---|
| Cumulative frequency | 12 | | | | 120 |

**[2]**

**(b)** On the grid, draw a cumulative frequency diagram to show this information.




**[3]**

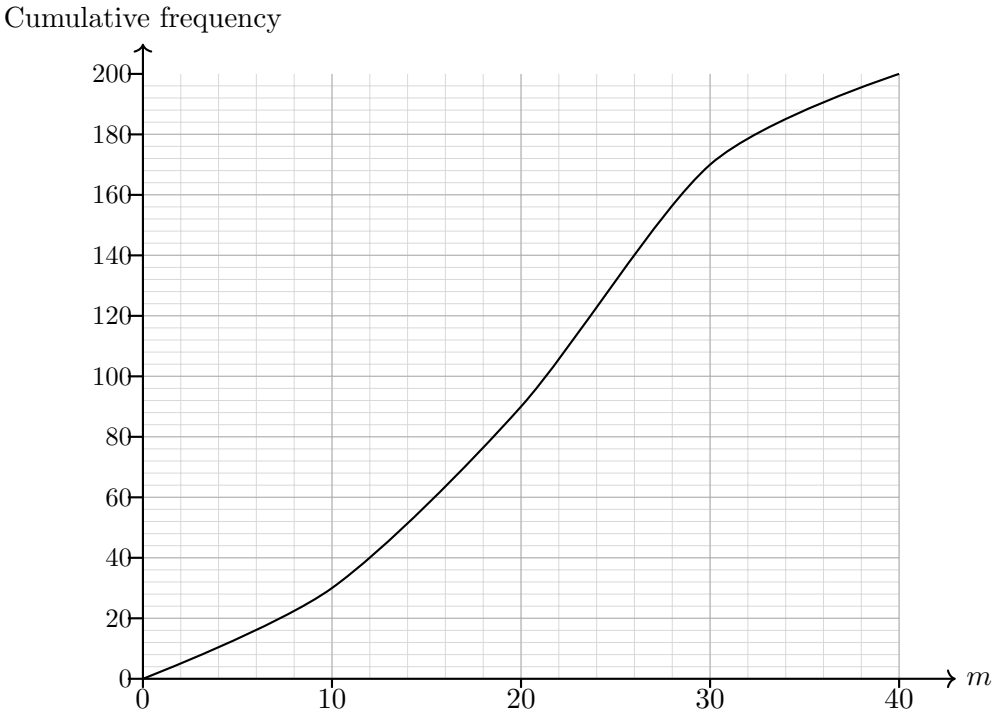
**(c)** Use your diagram to estimate the 60th percentile.

..... $g$ **[2]**

**(d)** Apples with a mass greater than 220 grams are classified as large.
Estimate the number of large apples.

..... **[3]**

**7** The cumulative frequency diagram shows information about the masses, $m$ grams, of 200 parcels.




**(a)** Find an estimate for the 80th percentile.

..... grams **[2]**

**(b)** Use the cumulative frequency diagram to complete the frequency table.

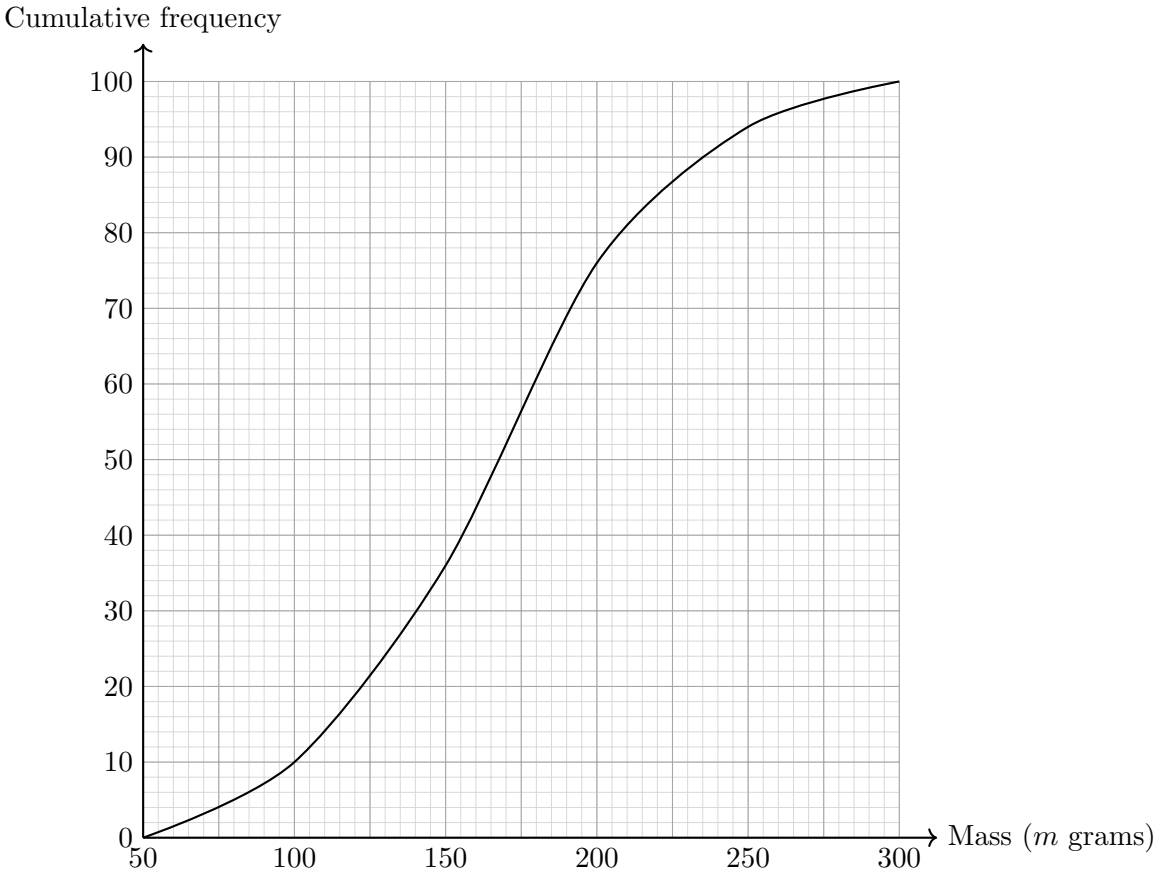
| Mass ($m$ grams) | $0 < m \leq 10$ | $10 < m \leq 20$ | $20 < m \leq 30$ | $30 < m \leq 40$ |
|---|---|---|---|---|
| Frequency | | | | |

[2]

**(c)** Calculate an estimate of the mean mass.

..... grams [4]

**8** The cumulative frequency curve shows information about the masses, $m$ grams, of 100 potatoes.




**(a)** Use the curve to estimate the number of potatoes with a mass greater than 220 grams.

..... **[3]**

**(b)** The curve was drawn using grouped data.
Find the values of $p$, $q$, and $r$.

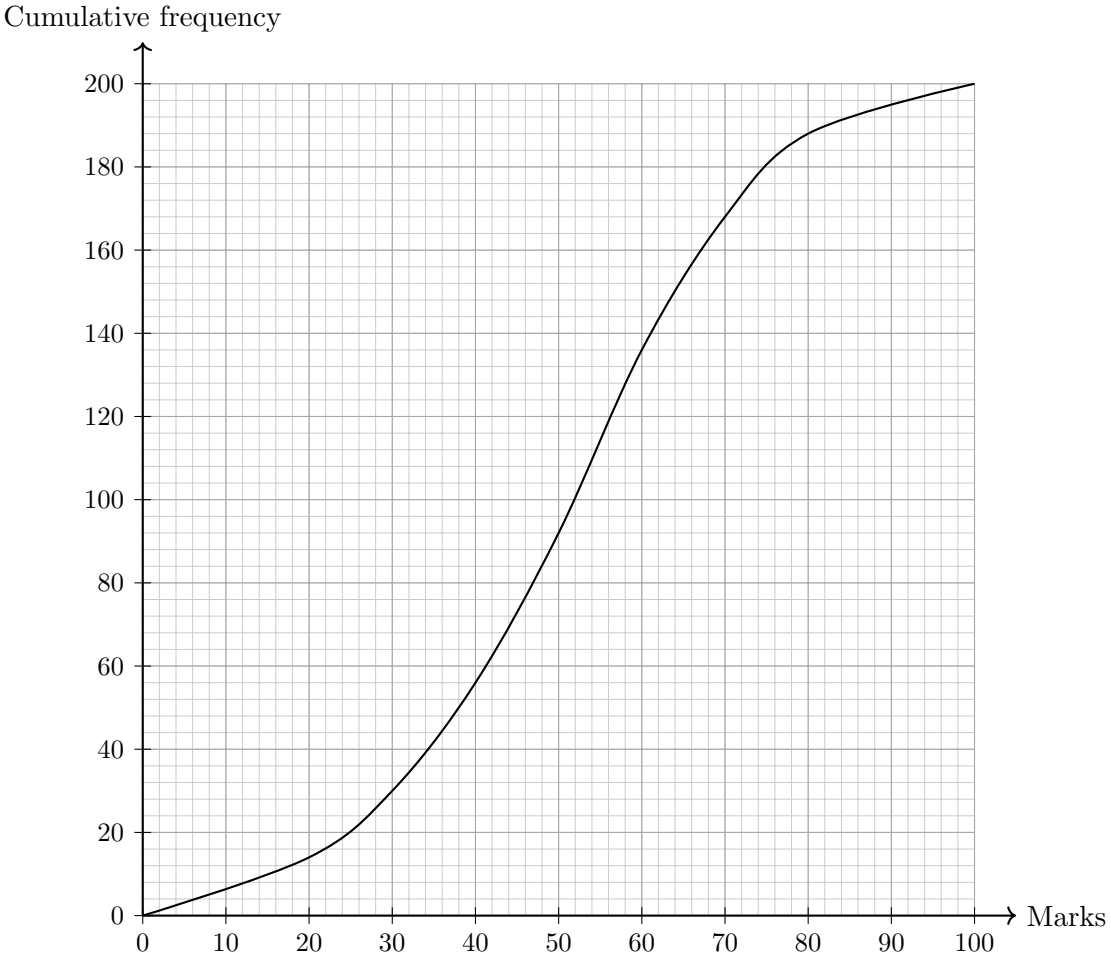
| Mass ($m$ grams) | $50 < m \leq 100$ | $100 < m \leq 150$ | $150 < m \leq 200$ | $200 < m \leq 250$ | $250 < m \leq 300$ |
|---|---|---|---|---|---|
| Frequency | $p$ | $q$ | 40 | $r$ | 6 |

$p =$ ..... , $q =$ ..... , $r =$ ..... **[3]**

**(c)** Calculate an estimate of the mean mass of the potatoes.

..... $g$ **[3]**

**9** The cumulative frequency diagram shows the distribution of marks scored by 200 candidates in an examination.




**(a)** Use your graph to estimate the interquartile range.

..... **[2]**

**(b)** 15% of the candidates score less than $m$ marks.
Find the value of $m$.

$m =$ ..... **[2]**

**(c)** Candidates scoring more than $P$ marks are awarded a Distinction.
Given that 30 candidates are awarded a Distinction, find the value of $P$.

$P =$ ..... **[2]**

**(d)** Candidates who score less than 45 marks have to take a supplementary test.
Calculate the percentage of candidates who do NOT have to take this test.

..... % **[4]**

**1** **(a)** 32 to 34 **[B1]**

**(b)** Reading Upper Quartile as 40 and Lower Quartile as 23 to 24 **[M1]**

16 to 17.5 **[A1]**

**(c)** $120 - 100$ (or 120 – their reading at $t = 45$) **[M1]**

18 to 22 **[A1]**

**[Back to Question]** *[Total Marks: 5]*

**2** **(a)** **(i)** 24 **[B1]**

**(ii)** Reading Lower Quartile = 15 and Upper Quartile = 32.5 (Accept 32 to 33) **[M1]**

17.5 (Accept 17 to 18) **[A1]**

**(b)** Reading Cumulative Frequency at 45 km = 56 (Accept 55 to 57) **[M1]**

$60 - 56$ **[M1]**

4 (Accept 3 to 5) **[A1]**

**[Back to Question]** *[Total Marks: 6]*

**3** **(a)** 6 points plotted correctly at the upper boundaries **[P2]**

(Allow P1 for 4 or 5 points plotted correctly)

A smooth curve drawn passing through all their points **[C1]**

**(b)** Reading from cumulative frequency 40 **[M1]**

$26 \leq$ answer $\leq 28$ **[A1]**

**[Back to Question]** *[Total Marks: 5]*

**4** **(a)** $90, 160, 188, 200$ **[B2]**

(B1 for 2 or 3 correct values)

**(b)** 5 points plotted correctly from their table **[B2]**

(B1 for 3 or 4 points plotted correctly)

Smooth curve drawn through their points **[B1]**

**(c)** Reading from curve at cumulative frequency = 140 **[M1]**

46 to 48 **[A1]**

**[Back to Question]** *[Total Marks: 7]*

**5** **(a)** Reading from cumulative frequency 60 **[M1]**

$22 \leq$ answer $\leq 24$ **[A1]**

**(b)** Attempt to read from cumulative frequencies 90 (UQ) and 30 (LQ) **[M1]**

Lower quartile $\approx 15.5$ and Upper quartile $\approx 29.5$ seen **[M1]**

$12 \leq$ answer $\leq 16$ **[A1]**

**(c)** Attempt to read at $d = 35$ (approx 105) **[M1]**

120 − their reading **[M1]**

$13 \leq$ answer $\leq 17$ **[A1]**

**[Back to Question]** *[Total Marks: 8]*

**6** **(a)** 46, 92, 112 **[B2]**

(Allow [B1] for 2 correct values)

**(b)** Points correctly positioned horizontally and vertically starting at $(50, 0)$ **[B1]**

All 5 points plotted correctly at class upper boundaries (Tolerance ± half small square) **[B1]**

Smooth curve drawn through all 6 points **[B1]**

**(c)** Finding target Cumulative Frequency = 60% of 120 = 72 **[M1]**

178 (Accept 176 to 180) **[A1]**

**(d)** Reading Cumulative Frequency at $m = 220$ is 100 **[M1]**

120 − 100 **[M1]**

20 (Accept 18 to 22) **[A1]**

**[Back to Question]** *[Total Marks: 10]*

**7** **(a)** $0.8 \times 200 = 160$ **[M1]**

$28 \leq$ answer $\leq 29$ **[A1]**

**(b)** Frequencies 30, 60, 80, 30 correct **[B2]**

(Allow B1 for 2 or 3 correct frequencies)

**(c)** Using midpoints 5, 15, 25, 35 **[M1]**

$\Sigma fx = 30 \times 5 + 60 \times 15 + 80 \times 25 + 30 \times 35$ (using their frequencies) **[M1]**

their $4100 \div 200$ **[M1]**

20.5 **[A1]**

**[Back to Question]** *[Total Marks: 8]*

**8** **(a)** Reading from curve at $m = 220$: 81 to 85 **[M1]**

100 − their reading **[M1]**

15 to 19 **[A1]**

**(b)** $p = 10$ **[B1]**

$q = 26$ **[B1]**

$r = 18$ **[B1]**

**(c)** Use of correct midpoints: $75, 125, 175, 225, 275$ **[M1]**

$\sum f \times m = 10 \times 75 + 26 \times 125 + 40 \times 175 + 18 \times 225 + 6 \times 275 = 16700$ **[M1]**

$16700 \div 100 = 167$ **[A1]**

**[Back to Question]** *[Total Marks: 9]*

**9** **(a)** Reading Lower Quartile = 38 (Accept 37 to 39) and Upper Quartile = 64 (Accept 63 to 66) **[M1]**

26 (Accept 24 to 29) **[A1]**

**(b)** Finding Cumulative Frequency = 15% of 200 = 30 **[M1]**

30 (Accept 29 to 31) **[A1]**

**(c)** Finding target Cumulative Frequency = 200 − 30 = 170 **[M1]**

71 (Accept 70 to 72) **[A1]**

**(d)** Reading Cumulative Frequency at 45 marks = 74 (Accept 72 to 76) **[M1]**

Number of candidates NOT taking test = 200 − 74 = 126 **[M1]**

$\frac{126}{200} \times 100\%$ **[M1]**

63 (Accept 62 to 64) **[A1]**

**[Back to Question]** *[Total Marks: 10]*